5.1 Introduction and Definitions (Easy Explanation)

The **biggest difficulty** in simulation is to check whether the simulation model is really correct and whether it truly represents the real system. This is called **model validity**. This chapter explains how to build **valid** and **credible** simulation models and how much **detail** a model should have (not too simple, not too complex). The ideas come from research papers, real consulting projects, many years of teaching and experience, and more than **40 examples** are used to explain the ideas.

This chapter mainly explains **three key terms** which are very important for exams ✅:

- **Verification**
- **Validation**
- **Credibility**

1. Verification (Very Simple Meaning)

Verification means checking whether the computer program is written correctly or not. In simple words, did we correctly convert the **model assumptions** into a **computer program**? Is the program free from coding errors (bugs)? So, **Verification = "Are we building the model right?"**. Verification is difficult because large simulation programs have many logic paths and debugging them takes a lot of time and effort.

2. Validation (Very Important Concept)

Validation means checking whether the simulation model correctly represents the real system for the given purpose. So, **Validation = "Are we building the right model?"**

Important Ideas about Validation (Exam Points):

- **Valid model helps decision making**: If a model is valid, decisions made using the model are similar to decisions made using the real system.
- **Validation difficulty depends on the system**: Simple systems are easy to validate, for example, a small neighborhood bank. Future or complex systems are hard or impossible to fully validate, for example, a naval weapons system in 2025.
- **No model is perfectly valid**: A simulation model is always an approximation. Perfect validity does not exist. Making a model more accurate requires more time, money, and data. But very high accuracy may not be worth the cost. Famous quote (important): **"All models are wrong, but some are useful." – George Box**. Meaning: models cannot show everything, but they can still help in decision making.
- **A model is built for a specific purpose**: A model valid for one purpose may not be valid for another purpose. For example, a model for reducing waiting time may not be valid for cost estimation.
- **Use correct performance measures**: Validation should use the same measures that managers use to judge the system.

- **Validation must be done during model development**: Validation should not be done at the end, it should be done throughout the project. Example 5.1: A company spent $500,000 on a simulation and after finishing, they asked: *“Can you tell us in 5 minutes how to validate the model?”* This shows poor planning.

- **Re-validation is necessary**: If a model is used for a new purpose, it must be validated again because old data or assumptions may no longer be correct.

3. Credibility

A model is **credible** if managers and decision makers believe the model is correct. Important points: a model can be credible but not valid, a model can be valid but not credible, and sometimes credible models are not used due to politics or budget issues.

How to Increase Model Credibility:

- Manager understands and agrees with assumptions
- Model is verified and validated
- Manager is involved in the project
- Model developers are well-known
- The model has good animation

VV&A (Verification, Validation, and Accreditation)

What is Accreditation? Accreditation means official approval that a model is acceptable for a specific purpose. Used mainly by the U.S. Department of Defense (DoD). Accreditation is needed when decisions involve huge money or human lives.

What is checked for Accreditation? Verification and validation results, model credibility, model history and past use, data quality, documentation quality, and known limitations of the model.

Validation vs Output Analysis (Very Important Difference)

Validation checks how close the model is to the real system. **Output Analysis** is statistical analysis of simulation results and deals with run length, warm-up period, and number of runs.

Simple Error Explanation: To estimate the real system mean, two types of errors exist:

1. Error due to model not matching system → Validation problem
2. Error due to randomness in simulation → Output analysis problem

For good results, both validation and output analysis are required.

One-Line Exam Summary

- **Verification:** Are we coding the model correctly?
- **Validation:** Does the model represent the real system?

- **Credibility:** Do people believe the model?
- No model is perfect, but useful models help decisions
- Validation and output analysis are both necessary

5.3 Verification of Simulation Computer Programs

Verification means checking whether the simulation computer program is written correctly or not. In simple words, it answers questions like: Are there coding mistakes (bugs)? Does the program work the way the model designer intended? Verification ensures that the program correctly implements the assumptions and logic of the simulation model. This section explains **eight techniques** commonly used to debug simulation programs, with examples.

Technique 1: Write and Debug in Parts (Modules)

A very large simulation program is hard to write and debug at once, so the correct approach is to develop the program in **small parts or modules**. Start by writing the **main program** and a few important subprograms. For the missing parts, use **dummy or stub programs**. Gradually, add more details. This method allows you to test and debug each module step by step, avoiding frustration when a large program fails.

Example 5.7 (Bank Model): First, write a bank model **without customer jockeying**. Once it works correctly, **add jockeying behavior**. This ensures each part is correct before increasing complexity.

Technique 2: Program Review by Multiple People

Programmers can easily **miss their own mistakes**, so reviewing the program by multiple people is helpful. In a **structured walk-through**, all team members sit together and check each line of code carefully. The team only moves to the next line if everyone agrees that the previous line is correct. This process improves **error detection** significantly.

Technique 3: Test with Different Input Values

Running the simulation with **different input parameter values** helps ensure that the program behaves correctly under various conditions. If the theoretical or exact results are known, compare the simulation output with these values.

Example 5.8 (Queue System): For a queue with **s servers**, the theoretical server utilization is $\lambda / (s\mu)$. If the simulation utilization is close to this value, the program is likely working correctly.

Technique 4: Use a Trace

A **trace** is a detailed record showing the system state, event list, state variables, and statistics after every event. Comparing trace outputs with **hand calculations** helps find logical errors, check extreme cases, and confirm that events occur in the correct order.

Types of Trace:

- **Batch-mode trace:** Produces a lot of output. Errors may be hard to find, and some important data may be missing.
- **Interactive debugger (better):** Pauses the simulation at any time, allows viewing and changing variable values, and forces certain errors to happen. Most modern simulation tools support interactive debugging, which is faster and more effective.

Example 5.9 (Single-Server Queue): A table shows the system state after initialization and after each arrival or departure. This helps verify correctness **step by step**.

Technique 5: Run Model Under Simple Conditions

Simplify the model so that the **true results are already known**, and then compare the simulation results with these known theoretical values.

Example 5.10 (Job-Shop Model): The original system is complex with no exact solution. By simplifying it to a **known queueing model**, the simulation results can be compared with theory. Close results indicate that the program is likely correct.

Example 5.11 (Cellular Network): Analytical solutions existed only for simple cases. Simulation results matched the analytical values, which increased confidence in the program.

Technique 6: Use Animation

Watching the simulation output visually helps detect **hidden errors** that are difficult to see in numerical results.

Example 5.12 (Traffic Simulation): The simulation looked correct numerically, but animation showed cars **colliding**, revealing programming errors that were not obvious otherwise.

Technique 7: Check Input Random Distributions

Compute **sample mean** and **sample variance** of random inputs and compare them with the desired or historical values. This ensures that random numbers and distributions are implemented correctly.

Example 5.13: Gamma and Weibull distributions are defined differently in some software. Checking statistical properties confirms correct implementation.

Technique 8: Use Simulation Packages

Commercial simulation software can reduce **manual programming**, but caution is required. New packages may contain **bugs**, some commands may not be well documented, and high-level macros may behave unexpectedly. Always verify results even when using packages.

Exam-Friendly Summary

Verification techniques include:

- Modular programming (writing in parts)
- Code review by multiple people

- Testing with different input values
- Using traces
- Simplifying the model to known conditions
- Animation
- Checking input random distributions
- Careful use of simulation software
- **Verification:** Checking whether the simulation program is coded correctly
- **Best debugging method:** Modular programming + trace + animation
- **Goal of verification:** Remove coding and logical errors

5.4 Techniques for Increasing Model Validity and Credibility

Main Idea

This section explains **methods to make a simulation model** valid → correctly represents the real system and credible → believable and accepted by managers. There are **six groups of techniques**. Here we explain **5.4.1: Collect High-Quality Information and Data**.

5.4.1 Collect High-Quality Information and Data on the System

Why is this important?

A simulation model is only as good as the **information and data** used to build it. 👉 **Bad data = wrong model = wrong decisions**.

Sources of Good Information

1. Conversations with Subject-Matter Experts (SMEs)

Who are SMEs?

People who **know the system very well**, for example operators, engineers, managers, and technicians.

Important Points

- A modeler **cannot work alone**
- No single person has all information
- The analyst must:

- Talk to **many SMEs**
- Identify the **correct SME** for each subsystem
- Avoid **biased information**

Best SMEs also understand **basic simulation ideas**. Even if no simulation is done, **collecting all system knowledge in one place is useful**. Since systems may change, the analyst must talk to SMEs **repeatedly**.

Example 5.14 (Manufacturing System)

Information should be collected from machine operators, engineers, maintenance staff, schedulers, managers, vendors, and blueprints.

Example 5.15 (Communication Network)

Relevant people include end users, network designers, technology experts, system administrators, maintenance staff, managers, and carriers.

2. Observation of the System

What does this mean?

If a **similar real system exists**, collect data from it. Data sources include historical records, time studies, and direct observation.

Two Important Rules for Data Collection

- **Data requirements must be clearly explained**
  - Type of data
  - Format
  - Amount
  - Conditions
  - Why data is needed
- **Understand how the data was produced**
  - Do not treat data as just numbers

Common Problems with Data (Very Important for Exams)

Problem 1: Data is not representative

The data may **not match real conditions**. Example 5.16 shows that military field test data is not equal to real combat conditions because of different behavior and no battlefield smoke.

Problem 2: Data is in the wrong format

The data exists, but **cannot be used directly**. Example 5.17 shows that machine failure data measured in **clock time** cannot be used when the simulation needs **busy time**, because idle time causes errors.

Problem 3: Measurement or recording errors

Data may be rounded or poorly recorded. Example 5.18 shows repair times rounded to the **nearest day**, so continuous probability distributions cannot be fitted.

Problem 4: Biased data

People may change data for **self-interest**. Example 5.19 shows the maintenance department exaggerated machine reliability to appear efficient.

Problem 5: Inconsistent units

Different units cause serious modeling errors. Example 5.20 shows that the Army and Air Force use **short tons (2000 lb)** while the Navy uses **long tons (2200 lb)**.

3. Existing Theory

Why use theory?

Mathematical theory helps decide probability distributions and system behavior. For example, if customers arrive at a constant rate, interarrival times are **exponential** and arrivals follow a **Poisson process**.

4. Results from Similar Simulation Studies

Use results from **previous studies**, especially useful when modeling military systems and large complex systems.

5. Experience and Intuition of Modelers

Some systems **do not yet exist** and some system behaviors are unknown. Modelers use past experience, engineering judgment, and intuition. These assumptions should be **tested and improved later**.

One-Line Exam Answers

- **SMEs:** People with deep knowledge of the system
- **Main threat to validity:** Poor or biased data
- **Best data practice:** Understand how data was produced

5.4.2 Interact with the Manager on a Regular Basis

One of the **most important ways** to make a simulation model **useful and accepted** is that the modeler must talk to the manager regularly during the study. This greatly increases the chance that the model will be **used** and the results will be **trusted**.

Why Regular Interaction with the Manager is Important

1. The real problem may not be clear at the beginning

At the start, the exact problem may be **unclear**. As the study continues, the problem becomes **better understood**, and the manager may need to **change or refine the objectives**. Even the best model is useless if it solves the wrong problem.

2. Keeps the manager interested

Regular meetings keep the manager **engaged**, and an interested manager is more likely to **use the model**.

3. Improves model validity

Managers often know the system very well, and their knowledge helps make the model **more realistic**.

4. Increases model credibility

The manager understands the assumptions, the manager agrees with the assumptions, and ideally, the manager should **sign off** on key assumptions. This makes the manager feel, "This is a good model because I helped build it."

- Frequent interaction with managers is essential
- Helps define the right problem
- Improves validity and credibility
- Increases chance of real-world use

5.4.3 Maintain a Written Assumptions Document and Perform a Structured Walk-Through

Many simulation models fail because of poor communication, missing assumptions, and wrong assumptions. Writing everything down clearly **reduces mistakes** and **builds trust**.

What is an Assumptions Document?

It is a **written description** of how the system is modeled, what assumptions are made, and what is included and what is not. It is also called a **conceptual model**. It is **not an exact copy** of the real system, and it describes the system **only for the study's purpose**. It represents the **modeler's vision** of the system.

Who Should Be Able to Read It?

The document should be understandable by analysts, subject-matter experts (SMEs), and technically trained managers.

What Should the Assumptions Document Contain? (Very Important for Exams)

1. Overview

- Project goals,Problems to be studied,Model inputs,Performance measures

2. Process-flow or system diagram

- Shows how the system works visually

3. Subsystem descriptions

- Written in **bullet points.**Explains:Each subsystem,How subsystems interact

4. Simplifying assumptions

- What simplifications were made,Why they were made.Simulation is a simplification of reality.

5. Model limitations

- What the model **cannot do,**Where results should be used carefully

6. Data summaries

- Mean values,Histograms.Detailed statistics go in appendices.

7. Sources of information

- People,Books,Papers,Reports

Why is the Assumptions Document So Important?

It acts as a **blueprint** for coding the simulation, helps avoid misunderstandings, and increases credibility.

Structured Walk-Through (Very Important Concept)

What is a Structured Walk-Through?

A structured walk-through is a formal meeting where the modeler presents the assumptions **bullet by bullet**, and everyone must agree before moving on. Participants include SMEs, managers, and all key project members.

Why is it Needed?

SMEs are busy and may give incomplete answers, important details may be missed, and group discussion catches errors early.

Best Practices

- Hold it **before programming starts**
- Preferably at a **remote location** (no distractions)
- Send document beforehand
- But still do the meeting (reading alone is not enough)

Benefits

- Finds missing or wrong assumptions
- Improves validity
- Improves credibility
- Creates **ownership of the model**

Example 5.21 (Successful Walk-Through)

Nine people attended, 160 assumptions were reviewed, it took 5½ hours, errors were fixed, new assumptions were added, and everyone agreed the model was valid. All participants felt the model was **theirs**.

Example 5.22 (Problems Discovered)

The sponsor's assumptions were wrong, SMEs corrected them, and a second walk-through fixed issues. This shows the importance of having **all key people involved early**.

Important Fact (Exam-worthy) About **75% of simulation models** have **poor documentation**.

5.4.4 Validate Components of the Model Using Quantitative Techniques

Use **numbers, statistics, and tests** to check if parts of the model are correct.

Examples of Quantitative Validation Techniques

1. Checking probability distributions

- Fit distributions to data
- Use:Graphs,Goodness-of-fit tests (Chapter 6)

2. Checking whether data sets can be combined

- Use **Kruskal–Wallis test**,**Checks if different data sets come from the same population**

Example 5.23

Two machines looked identical, the test showed they behaved differently, and each machine needed its own distribution.

Sensitivity Analysis (Very Important)

What is Sensitivity Analysis?

Change one or more model inputs and see how much the output changes. This helps find **important factors**.

Factors Tested in Sensitivity Analysis

- Parameter values
- Choice of distribution
- Entity type
- Level of detail
- Which data are most important to collect

Example 5.24 (Parameter Value)

The parameter was 0.75, tested with 0.70 and 0.80, and if the output does not change much, the parameter is not critical.

Example 5.25 (Entity Choice)

One candy bar was too slow for the simulation, 150 candy bars were used as one entity, and the same results were obtained with a faster simulation.

Example 5.26 (Level of Detail)

The assembly area was simplified, the output changed only by 2%, and the extra detail was unnecessary. However, the simplified model **cannot** be used for all purposes.

Important Warnings

- Use **common random numbers** in sensitivity analysis
- Otherwise randomness may hide real effects

Multiple Factors

- Do **not** change one factor at a time blindly
- Use **statistical experimental design** (Chapter 12)
- Helps detect:Main effects,Interaction effects

5.4.5 Validate the Output from the Overall Simulation Model

What does this mean?

The **best way to check if a simulation model is correct** is to see whether its **results (output)** look **similar to the real system's results**. This is called **results validation**. If the model gives outputs close to the real system, then the model is **valid and trustworthy**.

Method 1: Compare with an Existing Real System

Idea (in simple words):

If a **real system already exists**, then the following steps are used:

- First, **build a simulation model of the existing system**
- Run the simulation
- **Compare simulation results with real system results**

If both results are **very close**, then the model is **valid** and the model can be trusted for future decisions. Then, the model can be **modified** to represent the **new or proposed system**. The more similar the old and new systems are, the more confidence we have in the model.

How do we compare results?

We compare using:

Numerical measures

- Sample **mean**

- Sample **variance**
- Sample **correlation**

Graphical methods

- Histograms
- Box plots
- Distribution curves
- Spider-web (radar) plots
- Dot plots

Example 5.27 (Machines in a factory)

A factory had **2 machines**, and management wanted to buy a **3rd machine**. First, the model was tested using **2 machines**, and simulation results were compared with real data.

Result

The throughput error was **very small**, and utilization error was acceptable, so the model was considered **valid**. When the model was run with **3 machines**, it showed that the extra machine was **not needed**. The big benefit was that the company saved **$1.4 million**.

Example 5.28 (Air Force bombers)

A simulation model was built for bomber availability, and real data from **9 months** was available. Simulation and real availability differed by **less than 3%**, which showed the model was **highly valid**.

Example 5.29 (Telecommunication switch)

A real switch was tested in a lab using artificial traffic, and a simulation model was tested with the **same traffic**. Performance results were **very close**, so the model was accepted as **valid**.

Example 5.30 (Missile accuracy)

What was done?

The following steps were carried out:

- 8 real missiles were tested
- 15 simulated missiles were run
- Impact points were compared using:

- Graphs,Mean,Variance,Miss distance

Miss distance formula

$$d = \sqrt{x^2 + y^2}$$

Conclusion

Simulation missiles had a **larger average miss distance** and **larger variation**, so the result was that the model was **not valid** for missile accuracy.

Extra Method: Turing Test

What is a Turing Test (simple)?

Experts such as engineers and managers are shown some **real system data** and some **simulation data**, and they are **not told** which is which.If experts can tell the difference:

- The model needs **improvement**
- Their feedback helps fix the model:

Comparison with Expert Opinion

Even if there is **no existing system**, you can still check a simulation model by asking **subject-matter experts (SMEs)** to review the results. Experts check if the **simulation results seem reasonable** compared to what they know about the system. This is called **face validity** — the model "looks valid" to experts.

Care: Experts shouldn't just be told the "correct answer." The model's purpose is to find results we **don't already know**.

Example 5.34 (US Air Force manpower model)

A model was built to show effects of personnel policies. Experts (Air Force analysts) reviewed results under baseline policy. They found **some discrepancies** with expected behavior. Modelers **corrected the errors**, and repeated testing was done. Finally, the model **closely matched real policy**.

Result: Validity and credibility of the model improved.

Comparison with Another Model

If another simulation model exists for the **same system or similar purpose**, we can compare results in the following ways:

- Compare numerical statistics (mean, variance, etc.)
- Compare graphical plots (histograms, box plots, etc.)
- Optionally, use **formal statistical tests** like confidence intervals

Warning: Two models may give similar results but still both could be wrong.

Example 5.35 (Defense supply simulation)

A new model was built to **replace an old model** for ordering supplies. First, old model checked: total orders for 1996 compared with real system → differed < 3% → old model valid. Then, new model checked: predicted orders for 1998 compared with old model → differed < 6% → new model reasonably valid.

Extra idea: It would be better to also check smaller groups of data (e.g., categories of stock items). Sometimes, errors in different categories cancel out in totals, hiding problems.

Exam-Friendly Summary

- **Face validity:** Experts check if results look reasonable.
- **Comparison with another model:** Compare outputs, graphs, or confidence intervals.
- Even if two models agree, check carefully — both could have errors.
- Small-scale checks (like categories of items) are useful to catch hidden errors.

5.4.6 Animation

- **Purpose:** Animation shows a visual representation of the simulation.
- **Benefits:**
  - Helps find **wrong assumptions** in the model.
  - Increases **credibility**, especially for managers or non-technical stakeholders.

Example 5.36

A candy packaging system was animated. An operations manager saw it and said, "That's my system!" → The model gained instant credibility.

Tip: Animation is especially useful when showing the model to people who don't understand simulation details.

5.6 Statistical Procedures for Comparing System vs. Model Data

Goal

Check if the simulation output matches real-world observations.

- Let:
  - $(R_1, R_2, ..., R_k)$ = real system observations
  - $(M_1, M_2, ..., M_l)$ = model output data

- **Challenge:** Standard tests (t-test, chi-square, KS test) assume IID data, but simulation and system outputs are often:
  - **Nonstationary** → distributions change over time
  - **Autocorrelated** → observations are related to each other
- **Better question:** Are the differences **big enough to affect decisions**, rather than asking if they are exactly equal?

5.6.1 Inspection Approach

- **Idea:** Compare simple statistics from real system and model (like mean, variance, correlation) **without formal tests**.
- **Graphical checks** (histograms, box plots) are also useful.
- **Problem:** Each statistic is like a **sample of size 1**, so it can be very **misleading** due to randomness.

Example 5.37 (My/My/1 queue)

System: (r = 0.6), Model: (r = 0.5), Arrival rate = 1. Output: delays (D_1, D_2, ...) for 200 customers. Compute sample mean delays: (\hat{X} = 0.87) (system), (\hat{Y} = 0.49) (model). Results vary a lot between experiments → sometimes looks like the model is fine, sometimes not.

Conclusion: Simple inspection can be misleading.

Correlated Inspection Approach (Better Method)

- **Idea:** Feed the **model with historical system inputs** (actual arrival times, service times).
- **Benefit:** Both system and model see **exactly the same inputs**, so output comparisons are more precise.
- Also called **trace-driven simulation**.
- More reliable for validating assumptions **other than probability distributions** (distributions are validated separately in Chapter 6). **Key Tip:** Using correlated inputs reduces the randomness effect and gives a clearer view of whether the model is valid.

Feature	Basic Inspection	Correlated Inspection
Input for model	Random/independent	Same as system
Covariance	~0	High (~0.99)
Variance of difference	Large	Small
Accuracy of estimate	Low	High
Reliability	Can mislead	More trustworthy

5.6.2 Confidence-Interval Approach Based on Independent Data

Purpose

Compare a simulation model with the real system **using independent data sets** from both. This approach works best when **many system and model observations** are available (e.g., lab systems). It is often not feasible in real-life military or manufacturing systems due to limited data.

Method

- Let:
  - X_j = random variable from the **system** (e.g., average delay, miss distance)
  - Y_j = same variable from the **model**
  - $mX = E(X_j)$, $mY = E(Y_j)$
- Define the difference:
$$
z = mX - mY
$$
- Construct a **confidence interval (CI)** for z.
- **Paired-t approach:** use if the same input data drives the system and model (correlated) → requires $m = n$
- **Welch approach:** use if independent observations → $m \ge 2, n \ge 2$

Why CI is Better than Hypothesis Test

The null hypothesis (H_0: $mX = mY$) is almost always false because the model is only an approximation. A confidence interval gives the **magnitude and direction** of the difference, not just whether it is "significant or not." It helps decide **practical significance**, meaning whether the difference is large enough to affect decision-making.

Practical significance: A difference is practically significant if it could **change conclusions** drawn from the model.

Examples

Example 5.41 (Paired-t, Five-Teller Bank)

$W_j = X_j - Y_j$ is the difference for experiment j. There are $m = n = 10$ experiments. The sample variance is $\hat{Var}(W) = 0.02$. The 90% confidence interval for z is
$$
W \pm t_{9,0.95} \sqrt{\hat{Var}(W)} = -0.69 \pm 0.26 = [-0.95, -0.43]
$$
The confidence interval does **not contain 0**, so the difference is statistically significant. Practical significance still requires SME judgment, because the difference may or may not matter in practice.

Example 5.42 (Welch Approach, Missile System)

The system has 8 test missiles (X_j) and the model has 15 simulated missiles (Y_j). The sample statistics are

$$
X = 159.95,\; Y = 183.50,\; S_X^2 = 355.75,\; S_Y^2 = 545.71
$$

The 95% confidence interval for ($z = mX - mY$) is

$$
z \pm t_{\hat{f},0.975} \sqrt{\frac{S_X^2}{8} + \frac{S_Y^2}{15}} = 223.55 \pm 18.97 = [-242.52, -24.58]
$$

The confidence interval does **not contain 0**, so the difference is statistically significant. Practical significance again requires SME evaluation.

Key Points

- CI approach handles **independent multiple observations** from system and model.
- CI gives **more information** than a simple hypothesis test.
- Paired-t = correlated inputs, Welch = independent inputs.
- Large amounts of data may be needed.
- CI approach **does not account for autocorrelation** in output processes.

5.6.3 Time-Series Approaches

Compare **model output** and **system output** using **single time series** from each. The advantage is that only **one data set** is needed per system/model. This approach captures the **autocorrelation structure** (correlation between successive observations) and avoids the large data requirements of the replication or confidence-interval approach.

Main Methods

- **Spectral-Analysis Approach**
  - Compute the **sample spectrum** (Fourier cosine transform of autocovariance).
  - Construct CI for the **difference of log-spectra** to compare autocorrelations.
  - **Drawbacks:**
    - Requires **covariance-stationary** output (rare in practice).
    - Mathematically complex.
    - Hard to link CI directly to model validity.
- **Parametric Time-Series Approach (Box-Jenkins)**

    - Fit a **parametric time-series model** to each output series.
    - Apply a **hypothesis test** to see if the models are statistically similar.
    - Limitation: hypothesis tests are less informative than confidence intervals.
- **Standardized Time-Series Approach (Chen & Sargent, 1987)**
    - Construct a **CI for the difference in steady-state means** of system and model.
    - Only **one output series needed** per system/model.
    - Assumes independence and some standard statistical conditions.

The key idea is that time-series approaches allow validation **even when only one run of system data exists**, and they give insight into the **temporal patterns** of the outputs.

5.6.4 Other Approaches

Methods

- **Regression-Based Hypothesis Test (Kleijnen et al., 1998)**
    - Tests if **model mean = system mean** and **model variance = system variance**.
    - Applicable to **trace-driven simulation** with at least 3 IID observations per set.
    - Properties (Type I error, power) are evaluated using queues like M/M/1.
- **Bootstrap-Based Hypothesis Test (Kleijnen et al., 2000, 2001)**
    - Distribution-free test to check if **model mean = system mean**.
    - Works for **trace-driven simulation**.
    - Requires ($n \geq 3$) system observations and ($s \cdot n$) model observations.
    - Statistical properties tested on queues like M/M/1.

The authors and the text **prefer confidence intervals** over hypothesis tests for practical validation.

Exam Tip

- **Time-series CI methods** are best when **autocorrelation matters** or only one set of output is available.
- **Regression or bootstrapping tests** are useful for trace-driven models with multiple IID observations.
- Always consider **practical significance**, not just statistical significance.